PROJECT SYNOPSIS - SUBMISSION

HCLTech Career Shaper Programme | III Year Industry Project | 2025-26

FIELD	DETAILS
Project Title	AI-Based Resume Screening and Candidate Shortlisting System
Project No.	34
Domain	Digital-Data-AI / AIML
Difficulty	Advanced
Duration	4 Weeks
Format	Individual Project
Learner Name	Tanishq Tyagi
Roll Number	2300320100262
College / Campus	ABES Engineering College, Ghaziabad (4th Year)
Mentor Name	Kathiresan
Submission Date	30 October 2026

GitHub repository: https://github.com/Tan-hue-123/ai-resume-shortlist

1. Problem Statement

Recruitment teams reviewing large resume volumes need to extract job-related evidence and compare it with changing job descriptions, but manual review is time-consuming and basic keyword filters can miss equivalent wording. This project implements an explainable screening assistant that accepts PDF/TXT resumes, matches role-relevant skills and text similarity, and presents an advisory score for human review rather than making a hiring decision.

2. Proposed Solution & Approach

The built prototype is a Python Flask application with Jinja/HTML/CSS pages, PDF/TXT intake, pdfplumber-based PDF text extraction, a role-specific skill dictionary, scikit-learn TF-IDF cosine similarity, and SQLite review records. Its live fit signal combines 65% detected skill coverage and 35% TF-IDF similarity; a score of 65 or higher creates a review-for-shortlist recommendation, while a human reviewer records the final decision and rationale. The repository also contains a 5,000-row synthetic feature dataset, EDA scripts, and a separate logistic-regression baseline that is not called by the live upload flow. spaCy is listed in the project brief but is not used by the current repository implementation.

3. System Architecture

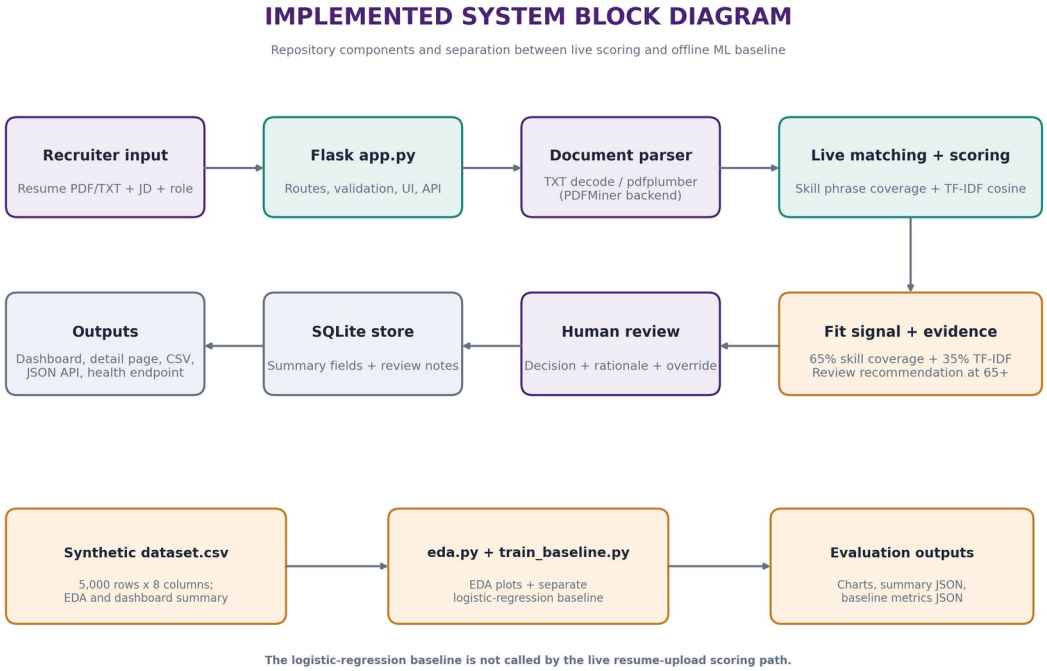


Figure S1. System block diagram showing the live Flask scoring path and separate offline synthetic-data analysis path.

3. System Architecture - Data Flow Diagram Level 0

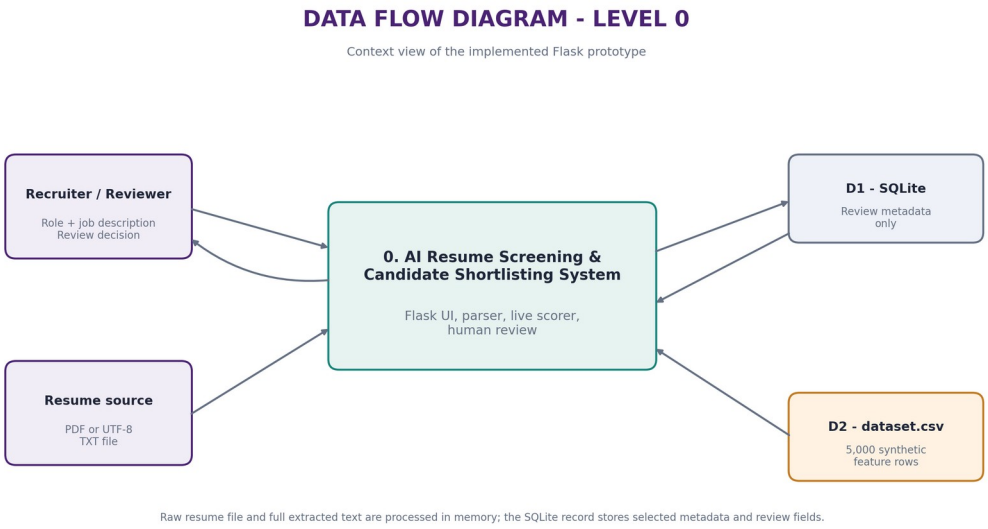


Figure S2. Data Flow Diagram Level 0, showing recruiter/resume inputs, system processing, and review/data stores.

4. Technology Stack

LAYER	TECHNOLOGY / IMPLEMENTED STATUS
Frontend / UI	Flask-rendered Jinja templates, HTML, CSS.
Backend / Logic	Python and Flask; app.py routes and validation.
Database	SQLite review store; CSV files for synthetic experiment data.
AI/ML Framework	scikit-learn TF-IDF and cosine similarity for live lexical matching; logistic regression for separate offline baseline. spaCy is not used in this version.
Document parsing	pdfplumber (uses pdfminer.six) for text-based PDF; UTF-8 TXT decoding.
Visualisation	Pandas and Matplotlib.
Hardware	Standard workstation; no special hardware required.

5. Dataset Used

ITEM	VALUE
Dataset file	data/dataset.csv
Rows / columns	5,000 rows / 8 columns
Job roles	10 synthetic role families
Missing cells / duplicate IDs	0 / 0
Shortlisted labels	1,316 (26.3%)
Key columns	job_role, years_exp, skills_matched, education_score, predicted_score
Additional data	None; no raw resume text, names, gender or age fields in the CSV.

The first 500 rows supplied for the project were retained and 4,500 synthetic feature rows were added to reach 5,000 total. The schema remains the eight columns specified in the brief. These data are synthetic and do not represent real candidate outcomes.

6. Key Results & Metrics

METRIC	RESULT	SCOPE / INTERPRETATION
Dataset completeness	5,000 x 8; 0 missing cells; 0 duplicate IDs	Schema/range check from eda.py.
Live scorer example	72.0 / 100; 7 of 7 selected skills	Direct scorer call on sample resume/JD; advisory output.
Scorer response time	2.45 ms median; 3.36 ms p95	150 in-process calls; excludes HTTP, uploads, PDF parsing and DB.
Dataset score-label correlation	r = 0.585	CSV predicted_score vs shortlisted; not live scorer validation.
Separate baseline accuracy	81.1%	Logistic-regression model on a stratified synthetic-data holdout; not used by app.
Separate baseline F1 / ROC-AUC	55.5% / 83.0%	Synthetic-only test split; not real-world validity.
Unit tests	5 passed / 5 total	Scoring utility, input validation, schema and SQLite review helper.

Business value achieved: the prototype demonstrates a repeatable way to compare resume text with job requirements, expose matched/missing skill evidence, and capture a human review decision. It is an academic triage demonstration only; no real hiring outcome or productivity gain has been measured.

7. Challenges Faced & How Resolved

CHALLENGE	RESOLUTION / LIMIT
The initial 500-row seed did not provide a large, balanced dataset.	Retained the seed and added 4,500 bounded synthetic rows across ten roles; preserved schema and documented synthetic provenance.
The project brief suggests spaCy, but the uploaded repository does not use it.	Documentation now accurately describes Python phrase matching and scikit-learn TF-IDF; spaCy is not claimed as implemented.
The live heuristic is not a trained predictive model.	Reported its formula, example output and direct-call timing separately; classifier metrics are labelled as offline synthetic baseline only.
No demographic data is included.	No group fairness statistics are claimed. A narrow name/pronoun perturbation check was run, with explicit limits; a proper audit requires governed group data.
Earlier drafts could not show the app running; UI figures were static template previews.	Flask app installed and run locally; report Section 8 now uses real browser screenshots, and upload, PDF parsing, review, CSV and API checks passed (Report Section 8.1).

8. References

[1] HCLTech Career Shaper Programme, “III Year Industry Project - Project Brief,” Project 34, Academic Year 2025-26 (supplied project brief).

[2] Flask Documentation, https://flask.palletsprojects.com/.

[3] pdfplumber Documentation, https://github.com/jsvine/pdfplumber.

[4] scikit-learn, “TfidfVectorizer,” https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html.

[5] scikit-learn, “Model evaluation: quantifying the quality of predictions,” https://scikit-learn.org/stable/modules/model_evaluation.html.

[6] Pandas Documentation, https://pandas.pydata.org/docs/.

[7] NIST, “AI Risk Management Framework,” https://www.nist.gov/itl/ai-risk-management-framework.

GitHub repository: github.com/Tan-hue-123/ai-resume-shortlist

9. Ethical Compliance Declaration

I confirm that:

- ☑ All dataset usage complies with the source licence.
- ☑ No personally identifiable information (PII) is stored without consent.
- ☑ AI tools used for code/writing are disclosed in Appendix A.
- ☑ This project complies with the DPDP Act 2023 where applicable.

Note: This declaration is reproduced from the supplied template and confirmed by the learner. The project uses synthetic data and fictional test resumes only; AI tool use is disclosed in Appendix A.

Learner Signature: _____ Date: _____

Learner: Tanishq Tyagi | Roll No.: 2300320100262 | Mentor: Kathiresan | ABES Engineering College, Ghaziabad (4th Year)

Appendix A. AI Tool Disclosure

The AI tools used while preparing this project are listed below, as required by the Ethical Compliance Declaration.

TOOL	USED FOR	HOW THE OUTPUT WAS CHECKED
ChatGPT (OpenAI)	Drafting and revising prototype code, documentation and report/synopsis text; generating the additional synthetic dataset rows.	Dataset checked by eda.py (schema, ranges, missing values, duplicate IDs); code covered by 5 unit tests.
Claude (Anthropic)	Reviewing the package against the project brief and synopsis template; running the Flask app and capturing the real screenshots in report Section 8; re-running eda.py, train_baseline.py and the unit tests; revising report and synopsis wording.	Reported metrics regenerated from the repository scripts; live-run checks recorded in report Section 8.1.

No real candidate data is used anywhere in the project: the dataset is synthetic and every test resume is fictional. Reported figures come from running the repository scripts and the live-run checks, not from an AI tool's own claims. AI-generated text and code were reviewed by the learner, who remains responsible for the submitted work.

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